Annex B

Cover Page for College to submit with Prelim Examination Paper

Qn/No	Topic Set	Paper Number	Answers	Marks
1	Inequalities	9740/01	$-4 \le x < -1$ or $x \ge 3$; $x \ge 3$	4
2	Binomial Series	9740/01	(i) $-2\left(1-\frac{1}{24}x-\frac{1}{576}x^2+.....\right)$ Expansion is valid for $\lvert x\rvert < 8$ (ii) 1.441	5
3	System of Linear Equations	9740/01	12 kg of chicken, 1 kg of mutton and 4 kg of beef	3
4	AP,GP & Sigma Notation	9740/01	$1 < a < 2$; $a=\frac{3}{2}$, $S_\infty = 3$	5
5	Functions	9740/01	(i) $f^{-1}: x \to 1-\sqrt{1+\ln(x+2)}$, $x > -1$ (ii) Least k = 2, $\left(1-\sqrt{1+\ln 3}, 0\right)$	6
6	Application of Differentiation	9740/01	$\frac{r^2\left(4+\sqrt{3}\right)}{104}$	5
7	Differentiation	9740/01	(a) $\frac{2}{x\sqrt{x^4-1}}$ (b) $y=\frac{1}{2}\left(x+\frac{1}{2}\ln(2)\right)+1$; $y=-\frac{1}{2}\left(x+\frac{1}{2}\ln(2)\right)-1$	8
8	Integration	9740/01	(a) $\frac{1}{2}e^{2x}\tan^{-1}\left(e^{-2x}\right)+\frac{1}{4}\ln\left(e^{4x}+1\right)+C$ (b) $-\frac{1}{2}\sqrt{1-4x-2x^2}-\frac{1}{\sqrt{2}}\sin^{-1}\frac{\sqrt{2}(x+1)}{\sqrt{3}}+C$	6
9	Curve Sketching	9740/01	(a) $\frac{dy}{dx}=1-\frac{b+4-2a}{(x-2)^2}$ (b) Largest possible k = 1	7
10	Complex Numbers	9740/01	(i) $\sqrt{153}$ (ii) $1.33 \le \arg(z-6) < \frac{\pi}{2}$	8
11	Transformation	9740/01		9

12	Mathematical Induction, Summation & Series	9740/01	(ii) $=\dfrac{1}{2(2n+1)}-\dfrac{1}{2(2n+3)}$ (iii) $\dfrac{1}{6}-\dfrac{1}{2(2N+3)}$ (iv) $\dfrac{1}{2}-\dfrac{1}{2(2N+1)}$	11
13	Vectors	9740/01	(i) $r \bullet \begin{pmatrix}1\\3\\2\end{pmatrix}=-2$ (ii) $\overrightarrow{ON}=\begin{pmatrix}-1\\1\\-2\end{pmatrix}$ (iii) $\underset{\sim}{r}=\begin{pmatrix}-5\\-11\\-10\end{pmatrix}+\lambda\begin{pmatrix}3\\5\\5\end{pmatrix},\lambda\in$ [illegible] (iv) $a=2$, $b=108$ or -116 (v) $a\neq 2,\ b\in\Re$	12
14	Application of Integration	9740/01	(aii) $25\left(\dfrac{\pi}{9}+\dfrac{\sqrt{3}}{24}-\dfrac{3}{16}\right)$ (b) 55.6	11
	Pure Maths			
1	Maclaurin's Series	9740/02	$1+x-\dfrac{1}{2}x^2+\cdots$; 0.010	5
2	Vectors	9740/02	(ii) 50.7°	6
3	Differential Equations	9740/02	(a) $x+y-\ln\lvert x+y+1\rvert=\dfrac{1}{2}(x+1)^2+c$ (bii) $x=8-Ae^{-\frac{5}{4}t}$; t = 1.11hrs (biii) $x=8-8e^{-\frac{5}{4}t}$ When x = 8 (i.e. he completes his 8 km jog) , $t\to\infty$ Model predicts infinite amount of time to complete jog, therefore model is NOT suitable.	9
4	Recurrence	9740/02	(i) $\alpha=\sqrt{k}$ (iii) $0<k<4$ (iv) $x_2=0.966$, $x_3=0.988$ (v)	10

<table>
<tr><td></td><td></td><td></td><td>From (ii) and (iii), as $x_1 = 0.9 < \sqrt{k} = 1$,
we have $x_n < x_{n+1} < \sqrt{k} = 1$ for all $n \in Z^+$
Thus $x_1 < x_2 < x_3 < \cdots < \sqrt{k} = 1$

The sequence $x_1, x_2, x_3, \ldots$ is an increasing sequence that converges to 1.</td><td></td></tr>
<tr><td>5</td><td>Complex Numbers</td><td>9740/02</td><td>(a) $w = 1 + i$; $z = 2i(1+i) - i = -2 + i$
(bi) $z = e^{\frac{2k\pi}{5}i}$, $k = 0, 1, 2, 3, 4$</td><td>10</td></tr>
<tr><td></td><td>Statistics</td><td></td><td></td><td></td></tr>
<tr><td>6</td><td>P & C</td><td>9740/02</td><td>(i) 2520
(ii) 1260; 182</td><td>7</td></tr>
<tr><td>7</td><td>Prob</td><td>9740/02</td><td>(aii) $\frac{21}{115}$ or 0.183
(b) Greatest value of n = 9</td><td>8</td></tr>
<tr><td>8</td><td>Sampling</td><td>9740/02</td><td>(i) There are a total of 400 employees.
For sample of 50, $\frac{400}{50} = 8$
That is, after randomly choosing an employee in the first group of 8, every 8th employee on the list is subsequently chosen. Hence the level 1 group in the marketing department which has 7 employees may not be represented.

(ii) A stratified sample is obtained by randomly choosing $\frac{x}{400} \times 50$ employees (rounded to the nearest integer) for each of the 12 strata above where x is the number of employees in each stratum.

<table>
<tr><td>Skill level</td><td>Level 1</td><td>Level 2</td></tr>
<tr><td>Male</td><td>45</td><td>126</td></tr>
<tr><td>Female</td><td>75</td><td>154</td></tr>
</table>

(iii) No of level 1 male employees chosen
= $\frac{45}{400} \times 50 = 5.625 \approx 6$
No of level 2 male employees chosen =
$\frac{126}{400} \times 50 = 15.75 \approx 16$</td><td>4</td></tr>
</table>

			No of level 1 female employees chosen = $\frac{75}{400}\times 50 = 9.375 \approx 9$ No of level 2 female employees chosen = $\frac{154}{400}\times 50 = 19.25 \approx 19$ After deciding the number chosen for each stratum, these employees are chosen using random sampling.	
9	Poisson	9740/02	(i) 0.422 (ii) 0.0568 (iii) Least n = 10	9
10	Binomial, Sampling	9740/02	(i) 0.0598 (ii) $a = 4.98$	6
11	Testing	9740/02	(i) $H_0 : \mu = 60$ $H_1 : \mu > 60$ (iii) There is a probability of 0.05 of concluding that the resident's claim is correct (justified) when in fact it is not correct (justified). (iv) $\overline{x} > 67.9$ (v) If the mean speed $\overline{x} = 65\ km/h$ <67.9, we do not reject H_0 and conclude that there is insufficient evidence to conclude that the average speed of vehicles is greater than $60\,km/h$.	9
12	Regression	9740/02	(i) r = 0.991 (ii) There is an increase of 0.15257 mm for each gram added. When there is no mass attached, the length of the string is 210.6 mm (iii) The string will break if too much weight is added. <u>OR</u> The string may lose its elasticity if too much weight is added and the nature of the relationship would change. (iv) 15.257 (v) 0.991	8
13	Normal	9740/02	(i) 0.819 (ii) 682 cents or $6.82 (iii) 0.557	9